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Project : Syria Tel analysis

Business Understanding

Overview:

Syriatel telecommunications company operating in Kenya offers a variety of plans to their customers, ranging from international plans, voicemail plans, etc. The company has experienced growth but has a challenge in retaining customers, as they opt out of their services (churn). Acquiring new customers is typically more expensive than retaining existing ones, so reducing churn is a high-priority business goal.

Problem Statement:

To determine whether we can predict which customers are likely to leave soon, so the company can intervene early.

Key Business Questions

- Are there patterns in customer behavior that signal churn?
- Which customers are most at risk of leaving soon?
- What factors (e.g., pricing, service usage, complaints) drive churn?

Metrics:

From a business perspective:

- Reduction in churn rate
- Increase in retained customers
- ROI of retention campaigns

From a model perspective:

- Accuracy - To know how well our model is performing
- Recall (very important) - Catch as many churners as possible
- Precision - Avoid wasting resources on customers who won't churn
- AUC-ROC - Overall model performance

Objectives

1. Reduce customer churn
2. Increase customer lifetime value
3. Improve retention strategies

Data Understanding

In [1]:

```
# Importing the relevant libraries
import numpy as np
import math
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import roc_auc_score, ConfusionMatrixDisplay, classification_report,
accuracy_score, confusion_matrix, roc_curve
from sklearn.preprocessing import OneHotEncoder, MinMaxScaler
from imblearn.over_sampling import SMOTE
from sklearn import tree
%matplotlib inline
```

In [2]:

```
# Loading the dataset
data = pd.read_csv("Data\Syria_tel.csv")
data.head()
```

<>:2: SyntaxWarning: invalid escape sequence '\S'

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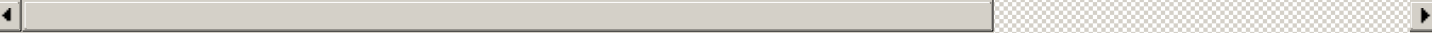
C:\Users\benbe\AppData\Local\Temp\ipykernel_26288\2405028064.py:2: SyntaxWarning: invalid escape sequence '\S'

```
data = pd.read_csv("Data\Syria_tel.csv")
```

Out[2]:

	state	account length	area code	phone number	international plan	voice mail plan	number vmail messages	total day minutes	total day calls	total day charge	...	total eve calls	total eve charge	total night minutes	total night calls	...
0	KS	128	415	382-4657	no	yes	25	265.1	110	45.07	...	99	16.78	244.7	91	...
1	OH	107	415	371-7191	no	yes	26	161.6	123	27.47	...	103	16.62	254.4	103	...
2	NJ	137	415	358-1921	no	no	0	243.4	114	41.38	...	110	10.30	162.6	104	...
3	OH	84	408	375-9999	yes	no	0	299.4	71	50.90	...	88	5.26	196.9	89	...
4	OK	75	415	330-6626	yes	no	0	166.7	113	28.34	...	122	12.61	186.9	121	...

5 rows x 21 columns

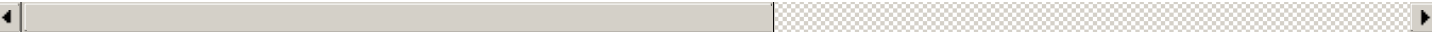


In [3]:

```
data.describe()
```

Out[3]:

	account length	area code	number vmail messages	total day minutes	total day calls	total day charge	total eve minutes	total eve calls	total eve charge
count	3333.000000	3333.000000	3333.000000	3333.000000	3333.000000	3333.000000	3333.000000	3333.000000	3333.000000
mean	101.064806	437.182418	8.099010	179.775098	100.435644	30.562307	200.980348	100.114311	17.083540
std	39.822106	42.371290	13.688365	54.467389	20.069084	9.259435	50.713844	19.922625	4.310668
min	1.000000	408.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	74.000000	408.000000	0.000000	143.700000	87.000000	24.430000	166.600000	87.000000	14.160000
50%	101.000000	415.000000	0.000000	179.400000	101.000000	30.500000	201.400000	100.000000	17.120000
75%	127.000000	510.000000	20.000000	216.400000	114.000000	36.790000	235.300000	114.000000	20.000000
max	243.000000	510.000000	51.000000	350.800000	165.000000	59.640000	363.700000	170.000000	30.910000



In [4]:

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3333 entries, 0 to 3332
Data columns (total 21 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   state                                3333 non-null   object
1   account length                       3333 non-null   int64
2   area code                           3333 non-null   int64
3   phone number                        3333 non-null   object
4   international plan                  3333 non-null   object
5   voice mail plan                     3333 non-null   object
6   number vmail messages               3333 non-null   int64
7   total day minutes                   3333 non-null   float64
8   total day calls                     3333 non-null   int64
9   total day charge                    3333 non-null   float64
10  total eve minutes                   3333 non-null   float64
11  total eve calls                     3333 non-null   int64
12  total eve charge                    3333 non-null   float64
13  total night minutes                 3333 non-null   float64
14  total night calls                   3333 non-null   int64
15  total night charge                  3333 non-null   float64
16  total intl minutes                  3333 non-null   float64
17  total intl calls                    3333 non-null   int64
18  total intl charge                   3333 non-null   float64
19  customer service calls              3333 non-null   int64
20  churn                              3333 non-null   bool
dtypes: bool(1), float64(8), int64(8), object(4)
memory usage: 524.2+ KB
```

Summary of the data

The data set has 21 columns of data(features) and 3333 data entries.

1. **state** - The state the customer comes from
2. **account length** - The account number of the customer
3. **area code** - The assigned code by the state indicating area zones
4. **phone number** - Unique identifier of customer
5. **international plan** - Shows whether customer has subscribed for the international plan
6. **voice mail plan** - Shows whether customer has subscribed for the voice mail plan
7. **number vmail messages** - The number of voice mail messages received by a customer
8. **total day minutes** - The duration of time spent by customer on calls during the day
9. **total day calls** - The number of calls made by customer during the day
10. **total day charge** - The charges for making a call during the day
11. **total eve minutes** - The duration of time spent by customer on calls during the evening
12. **total eve calls** - The number of calls made by customer during the evening
13. **total eve charge** - The charges for making a call during the evening
14. **total night minutes** - The duration of time spent by customer on calls during the night
15. **total night calls** - The number of calls made by customer during the night
16. **total night charge** - The charges for making a call during the night
17. **total intl minutes** - The duration of time spent by customer on international calls
18. **total intl calls** - The number of international calls made by customers
19. **total intl charge** - The charges for making an international call
20. **customer service calls** - Calls received by the customer care
21. **churn** - Likelihood of a customer stopping to use the service

EDA

In [5]:

```
# To convert the missing values in the data into nan type
data.replace('-', np.nan, inplace=True)
```

In [6]:

```
# Checking for null values in the data
data.isna().sum()
```

Out[6]:

```
state                0
account length       0
area code            0
phone number         0
international plan   0
voice mail plan      0
number vmail messages 0
total day minutes    0
total day calls      0
total day charge     0
total eve minutes    0
total eve calls      0
total eve charge     0
total night minutes  0
total night calls    0
total night charge   0
total intl minutes   0
total intl calls     0
total intl charge    0
customer service calls 0
churn                0
dtype: int64
```

In [7]:

```
# Checking for duplicates in the data
data.duplicated().sum()
```

Out[7]:

```
0
```

In [8]:

```
data["churn"].value_counts()
```

Out[8]:

```
churn
False    2850
True      483
Name: count, dtype: int64
```

In [9]:

```
data["total eve minutes"].value_counts()
```

Out[9]:

```
total eve minutes
169.9    9
167.2    7
180.5    7
201.0    7
161.7    7
..
75.3     1
293.9    1
232.8    1
173.2    1
265.9    1
Name: count, Length: 1611, dtype: int64
```

In [10]:

```
data['total eve charge'].value_counts()
```

Out[10]:

```
total eve charge
14.25    11
16.12    11
15.90    10
17.09     9
18.62     9
..
7.85     1
8.63     1
26.54     1
6.74     1
22.60     1
Name: count, Length: 1440, dtype: int64
```

In [11]:

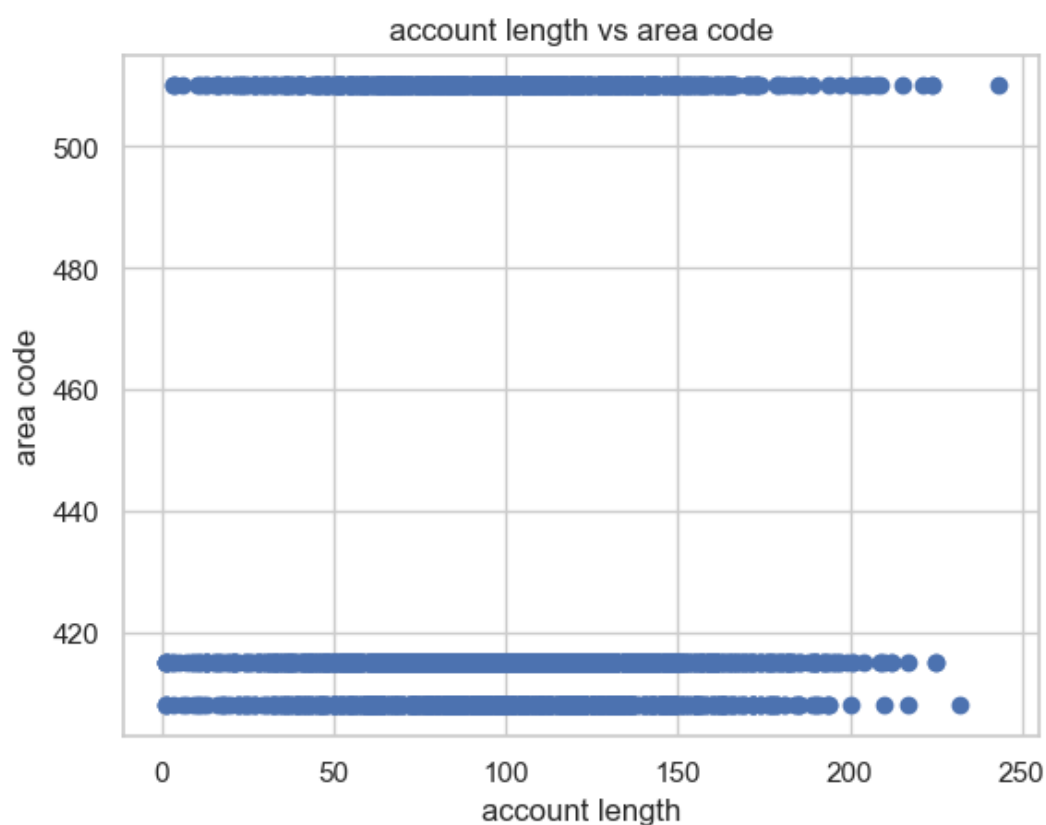
```
data['total day charge'].value_counts()
```

Out[11]:

```
total day charge
26.18     8
27.12     8
29.67     8
31.18     7
29.82     7
..
13.36     1
34.15     1
43.23     1
41.99     1
30.74     1
Name: count, Length: 1667, dtype: int64
```

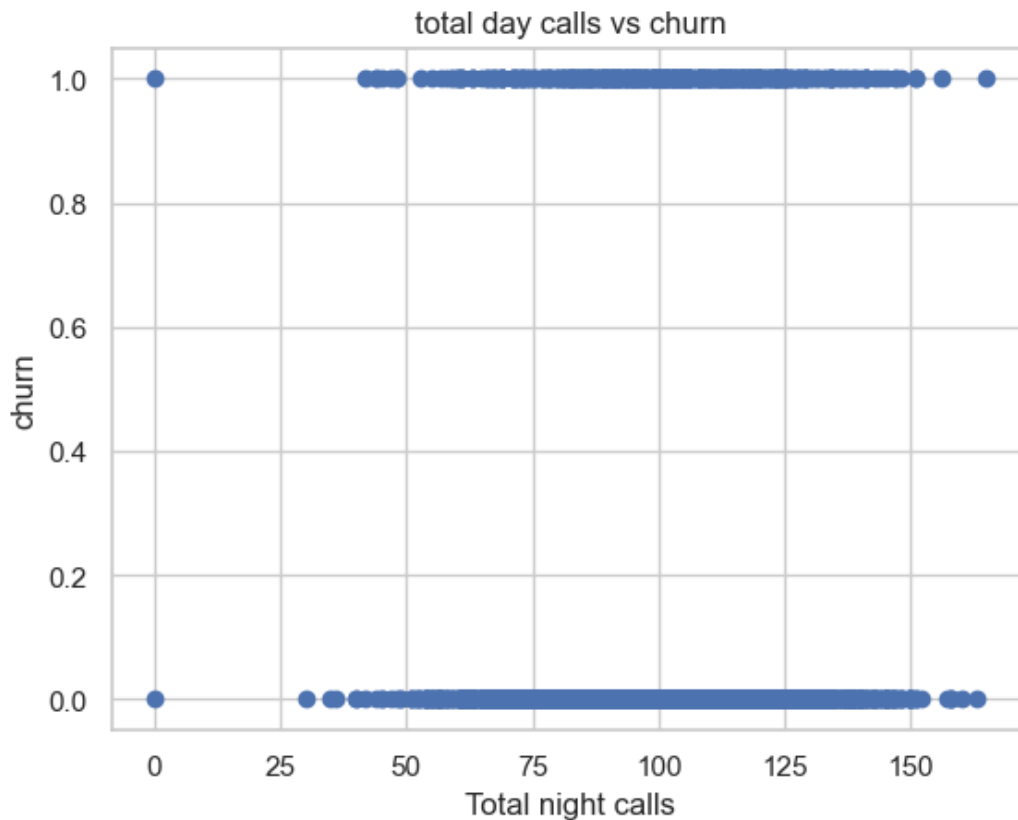
In [51]:

```
fig, ax = plt.subplots()
ax.scatter(data['account length'], data['area code'])
plt.title('account length vs area code')
ax.set_ylabel("area code")
ax.set_xlabel("account length")
plt.show()
```



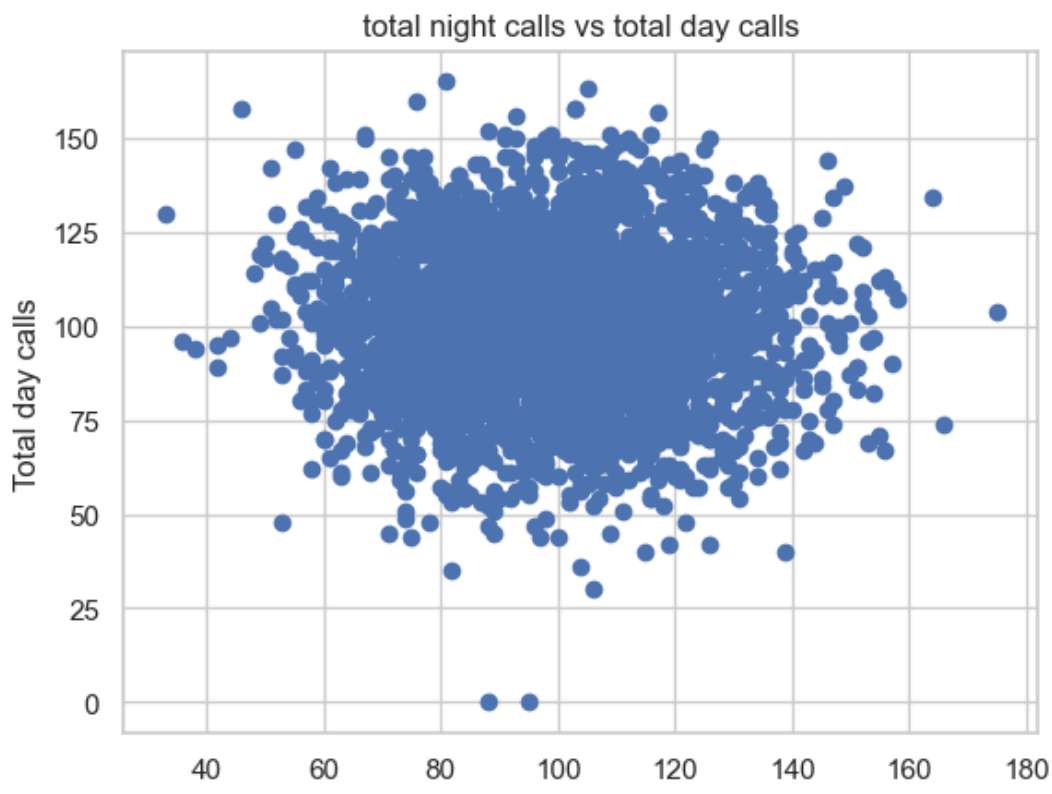
In [52]:

```
fig, ax = plt.subplots()
ax.scatter(data['total day calls'], data['churn'])
plt.title('total day calls vs churn')
ax.set_ylabel("churn")
ax.set_xlabel("Total night calls")
plt.show()
```



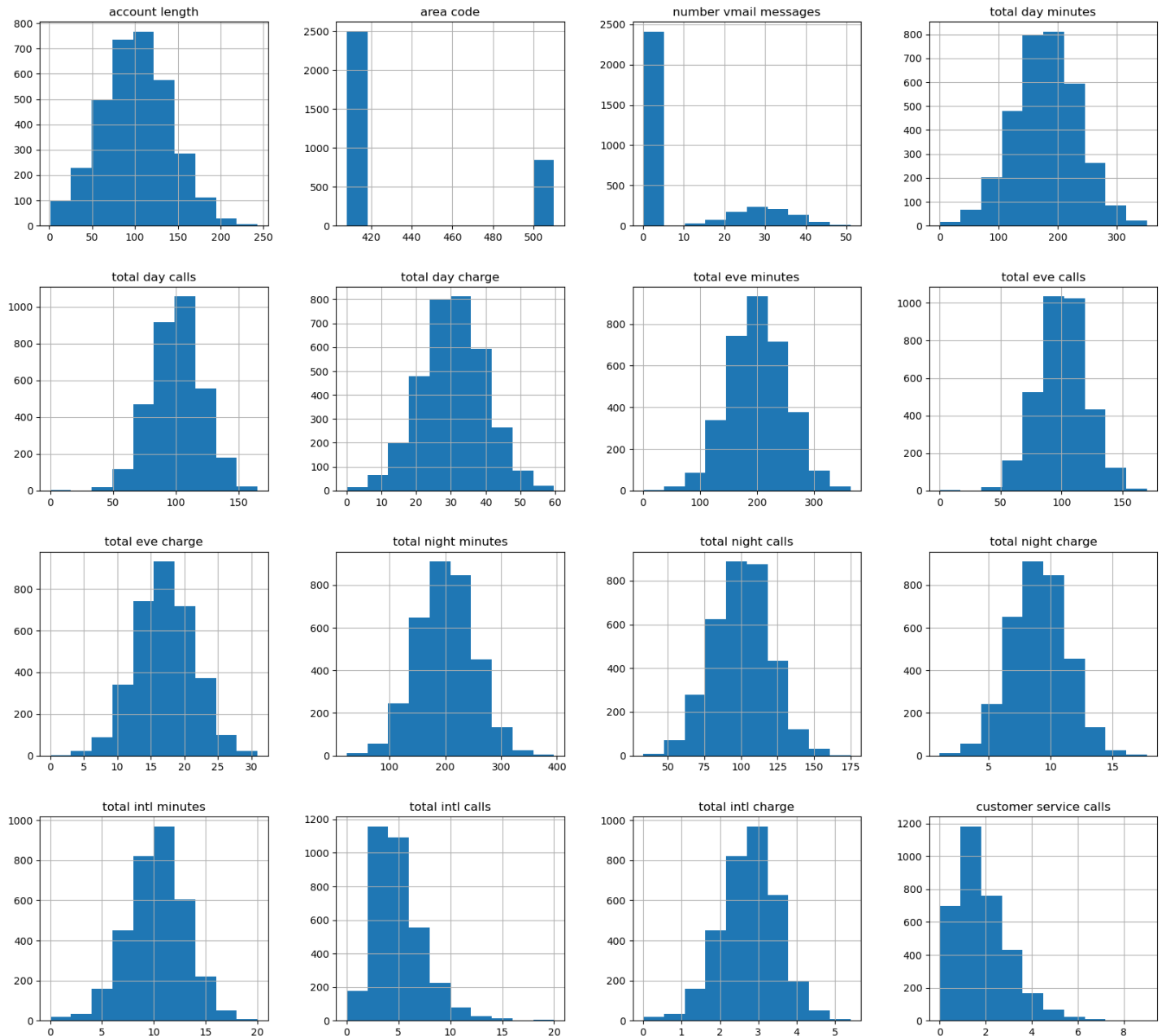
In [49]:

```
fig, ax = plt.subplots()
ax.scatter(data['total night calls'], data['total day calls'])
plt.title('total night calls vs total day calls')
ax.set_ylabel("Total day calls")
ax.set_xlabel("Total night calls")
plt.show()
```



In [15]:

```
data.hist(figsize = (20,18));
```



Discussion

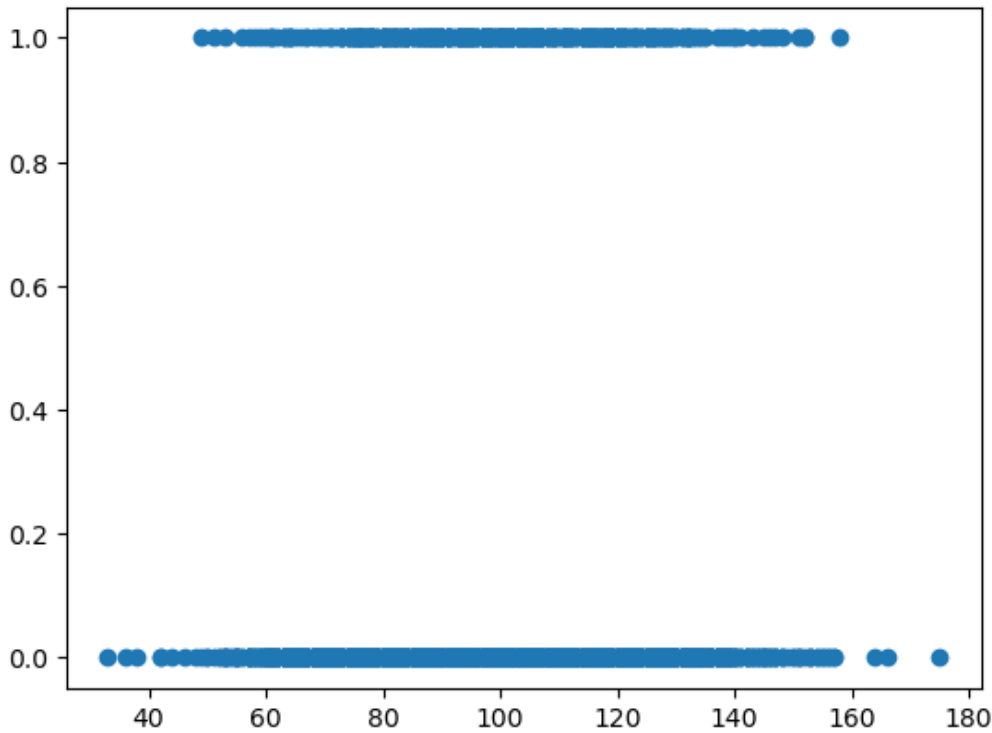
- The distributions show that most usage-related variables (e.g., total day/evening/night minutes and charges) are approximately bell-shaped, suggesting stable and consistent customer behavior across time periods. The close alignment between minutes and corresponding charges indicates strong linear relationships, which is useful for models like Logistic Regression.
- Variables such as number of voicemail messages and customer service calls are highly right-skewed, with many customers having low values and a few having very high counts. This suggests that extreme behaviors (e.g., frequent complaints) may be strong indicators of churn.
- The area code feature appears categorical with limited variation, implying it may have low predictive power unless combined with other regional insights. Similarly, account length is fairly normally distributed, indicating customers are spread across different tenure levels without strong imbalance.
- Call counts (day/evening/night) are moderately normally distributed, reflecting consistent calling patterns, while international calls and minutes are skewed toward lower values, indicating fewer users engage heavily in international usage.
- Overall, the dataset appears well-structured with mostly clean distributions, but skewed variables and potential outliers (especially in service calls and voicemail usage) should be carefully handled, as they could carry strong signals for identifying churn behavior.

In [16]:

```
fig, ax = plt.subplots()
ax.scatter(data['total night calls'], data['churn'])
```

Out[16]:

<matplotlib.collections.PathCollection at 0x1ec0da1af00>



In [17]:

```
data['phone number'].value_counts()
```

Out[17]:

```
phone number
382-4657      1
348-7071      1
389-6082      1
415-3689      1
379-2503      1
..
352-1127      1
368-1288      1
403-5279      1
397-9333      1
400-4344      1
Name: count, Length: 3333, dtype: int64
```

In [18]:

```
data["state"].value_counts()
```

Out[18]:

```
state
WV      106
MN       84
NY       83
AL       80
WI       78
OH       78
OR       78
WY       77
VA       77
CT       74
MI       73
```



```
ID      73
VT      73
TX      72
UT      72
IN      71
MD      70
KS      70
NC      68
NJ      68
MT      68
CO      66
NV      66
WA      66
RI      65
MA      65
MS      65
AZ      64
FL      63
MO      63
NM      62
ME      62
ND      62
NE      61
OK      61
DE      61
SC      60
SD      60
KY      59
IL      58
NH      56
AR      55
GA      54
DC      54
HI      53
TN      53
AK      52
LA      51
PA      45
IA      44
CA      34
Name: count, dtype: int64
```

In [19]:

```
data.drop(columns=['phone number', 'state'], inplace=True)
data.columns
```

Out[19]:

```
Index(['account length', 'area code', 'international plan', 'voice mail plan',
      'number vmail messages', 'total day minutes', 'total day calls',
      'total day charge', 'total eve minutes', 'total eve calls',
      'total eve charge', 'total night minutes', 'total night calls',
      'total night charge', 'total intl minutes', 'total intl calls',
      'total intl charge', 'customer service calls', 'churn'],
      dtype='object')
```

Data Preparation

In [20]:

```
# Splitting the columns based on variable types
categorical_columns = []
numerical_columns = []
for i in data.columns:
    if data[i].dtype == 'O':
        categorical_columns.append(i)
    else:
        numerical_columns.append(i)
```

```
print(f"Categorical columns:", categorical_columns)
print(f"Numerical columns:", numerical_columns)
```

```
Categorical columns: ['international plan', 'voice mail plan']
Numerical columns: ['account length', 'area code', 'number vmail messages', 'total day minutes', 'total day calls', 'total day charge', 'total eve minutes', 'total eve calls', 'total eve charge', 'total night minutes', 'total night calls', 'total night charge', 'total intl minutes', 'total intl calls', 'total intl charge', 'customer service calls', 'churn']
```

In [21]:

```
data['international plan'].value_counts()
```

Out[21]:

```
international plan
no      3010
yes      323
Name: count, dtype: int64
```

In [22]:

```
data['voice mail plan'].value_counts()
```

Out[22]:

```
voice mail plan
no      2411
yes      922
Name: count, dtype: int64
```

In [23]:

```
data['account length'].value_counts()
```

Out[23]:

```
account length
105      43
87       42
101      40
93       40
90       39
..
243       1
200       1
232       1
5         1
221       1
Name: count, Length: 212, dtype: int64
```

In [24]:

```
# Converting the categorical data to numerical data
encoder = OneHotEncoder(sparse_output=False)

one_hot_encoded = encoder.fit_transform(data[categorical_columns])

one_hot_df = pd.DataFrame(one_hot_encoded, columns=encoder.get_feature_names_out(categorical_columns))

df_encoded = pd.concat([data, one_hot_df], axis=1)

df_encoded = df_encoded.drop(categorical_columns, axis=1)
print(f"Encoded data : \n{df_encoded}")
```

```
Encoded data :
   account length  area code  number vmail messages  total day minutes \
0              128       415                    25          265.1
1              107       415                    26          161.6
2              137       415                     0          243.4
3               84       408                     0          299.4
..              ..       ..                      ..          ..
..              ..       ..                      ..          ..
```

4	75	415	0	166.7
...	...	...	...	...
3328	192	415	36	156.2
3329	68	415	0	231.1
3330	28	510	0	180.8
3331	184	510	0	213.8
3332	74	415	25	234.4
	total day calls	total day charge	total eve minutes	total eve calls \
0	110	45.07	197.4	99
1	123	27.47	195.5	103
2	114	41.38	121.2	110
3	71	50.90	61.9	88
4	113	28.34	148.3	122
...	...	...	...	...
3328	77	26.55	215.5	126
3329	57	39.29	153.4	55
3330	109	30.74	288.8	58
3331	105	36.35	159.6	84
3332	113	39.85	265.9	82
	total eve charge	total night minutes	... total night charge \	
0	16.78	244.7	...	11.01
1	16.62	254.4	...	11.45
2	10.30	162.6	...	7.32
3	5.26	196.9	...	8.86
4	12.61	186.9	...	8.41
...	...	...	...	...
3328	18.32	279.1	...	12.56
3329	13.04	191.3	...	8.61
3330	24.55	191.9	...	8.64
3331	13.57	139.2	...	6.26
3332	22.60	241.4	...	10.86
	total intl minutes	total intl calls	total intl charge \	
0	10.0	3	2.70	
1	13.7	3	3.70	
2	12.2	5	3.29	
3	6.6	7	1.78	
4	10.1	3	2.73	
...	...	...	...	
3328	9.9	6	2.67	
3329	9.6	4	2.59	
3330	14.1	6	3.81	
3331	5.0	10	1.35	
3332	13.7	4	3.70	
	customer service calls	churn	international plan_no \	
0	1	False	1.0	
1	1	False	1.0	
2	0	False	1.0	
3	2	False	0.0	
4	3	False	0.0	
...	...	...	...	
3328	2	False	1.0	
3329	3	False	1.0	
3330	2	False	1.0	
3331	2	False	0.0	
3332	0	False	1.0	
	international plan_yes	voice mail plan_no	voice mail plan_yes	
0	0.0	0.0	1.0	
1	0.0	0.0	1.0	
2	0.0	1.0	0.0	
3	1.0	1.0	0.0	
4	1.0	1.0	0.0	
...	...	...	...	
3328	0.0	0.0	1.0	
3329	0.0	1.0	0.0	
3330	0.0	1.0	0.0	
3331	1.0	1.0	0.0	
3332	0.0	0.0	1.0	

[3333 rows x 21 columns]

In [25]:

```
df_encoded.head()
```

Out[25]:

	account length	area code	number vmail messages	total day minutes	total day calls	total day charge	total eve minutes	total eve calls	total eve charge	total night minutes	...	total night charge	total intl minutes	total intl calls	total intl charge	cus s
0	128	415	25	265.1	110	45.07	197.4	99	16.78	244.7	...	11.01	10.0	3	2.70	
1	107	415	26	161.6	123	27.47	195.5	103	16.62	254.4	...	11.45	13.7	3	3.70	
2	137	415	0	243.4	114	41.38	121.2	110	10.30	162.6	...	7.32	12.2	5	3.29	
3	84	408	0	299.4	71	50.90	61.9	88	5.26	196.9	...	8.86	6.6	7	1.78	
4	75	415	0	166.7	113	28.34	148.3	122	12.61	186.9	...	8.41	10.1	3	2.73	

5 rows x 21 columns

Checking for Outliers

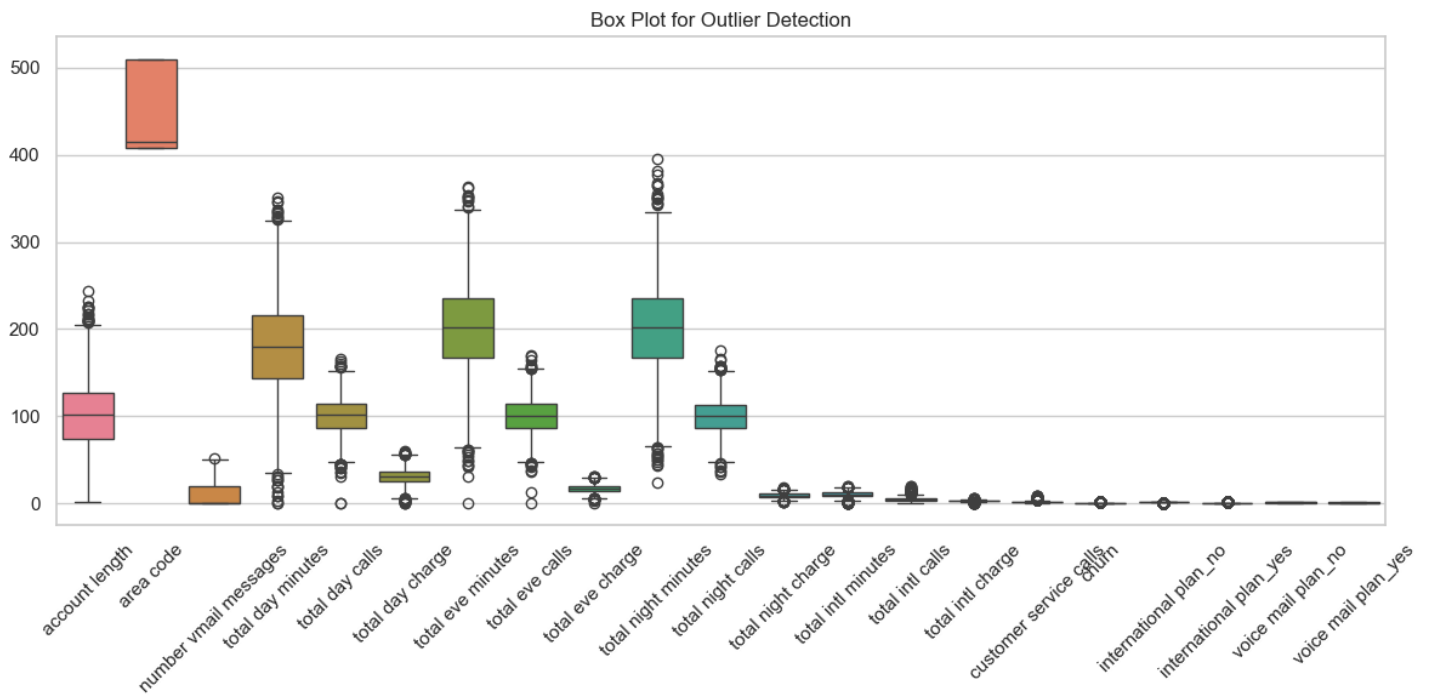
In [26]:

```
# Set style for better visuals
sns.set(style="whitegrid")

# Plot boxplots for all numeric columns
plt.figure(figsize=(12, 6))
sns.boxplot(data=df_encoded)

# Rotate x-axis labels for readability
plt.xticks(rotation=45)

plt.title("Box Plot for Outlier Detection")
plt.tight_layout()
plt.show()
```



Discussion

- The box plot shows that several continuous variables, especially total day, evening, and night minutes and

- The box plot shows that several continuous variables, especially total day, evening, and night minutes and charges, contain noticeable outliers, indicating some customers have unusually high usage.
- The box plot for total evening minutes has a higher average compared to the other minutes indicating more calls occur at night than other times (day, evening) which may be due to the lower night charges average compared to other times (day, evening)
- Customer service calls and number of voicemail messages also exhibit outliers, suggesting a subset of customers with extreme behavior.
- Features like area code and binary plan indicators show little variation and minimal outlier presence.
- The presence of outliers suggests the need for robust modeling or potential treatment (e.g., capping or transformation).
- Overall, variability is highest in usage-related features, which may strongly influence churn predictions.

Bivariate Analysis

In this section, we will perform bivariate analysis to examine the relationship between the target variable - churn and the other numeric and continuous features in the data. We will use scatter plots to show the direction, strength, and shape of the relationship between two numeric variables. This will help us understand how one variable affects or is affected by another variable and identify any patterns or trends that may exist.

In [27]:

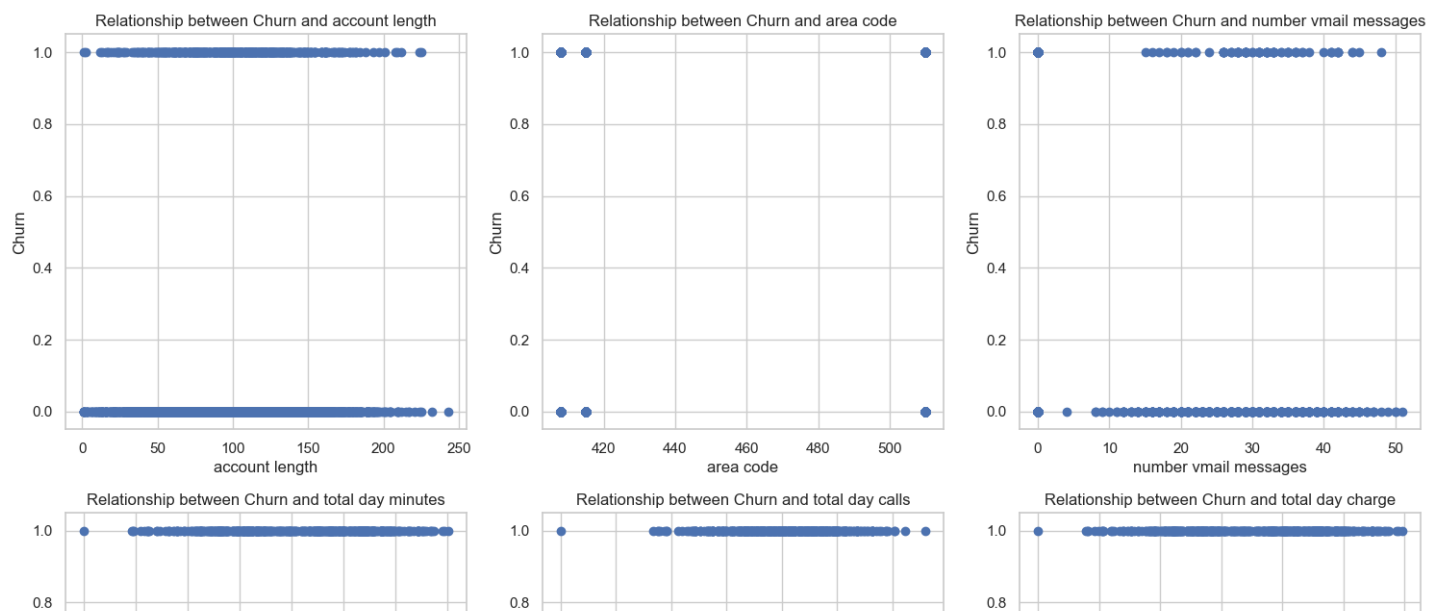
```
# Examine the relationships between 'churn' and several independent variables through scatter plots
variables = df_encoded.drop('churn', axis=1)
num_rows = math.ceil(len(variables.columns) / 3) # Calculate the number of rows for the subplots

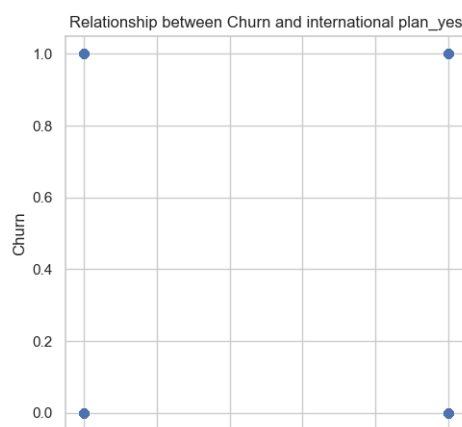
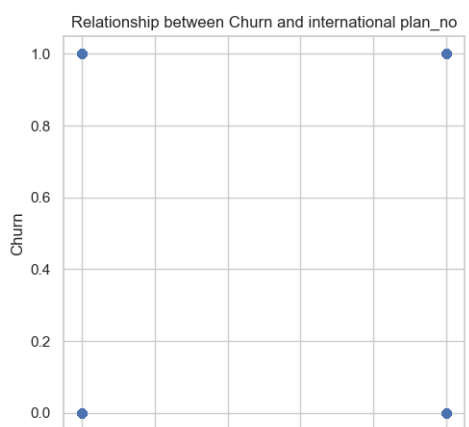
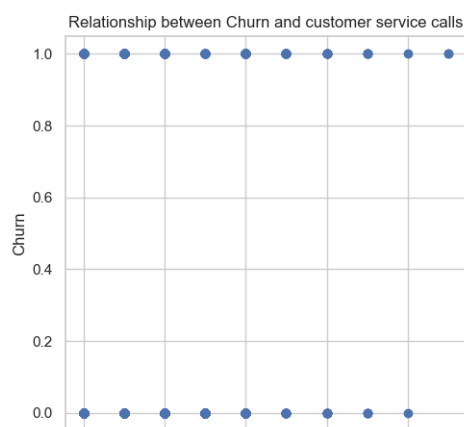
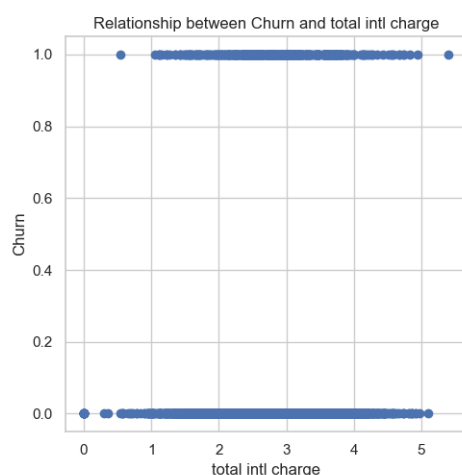
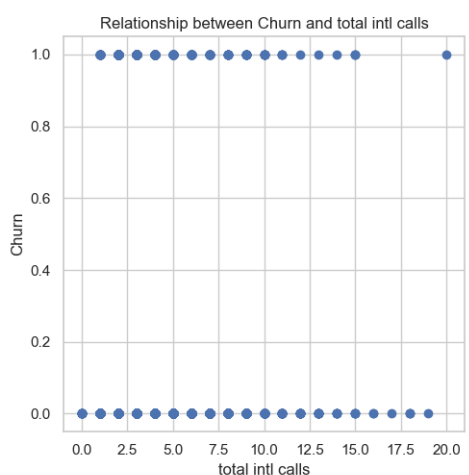
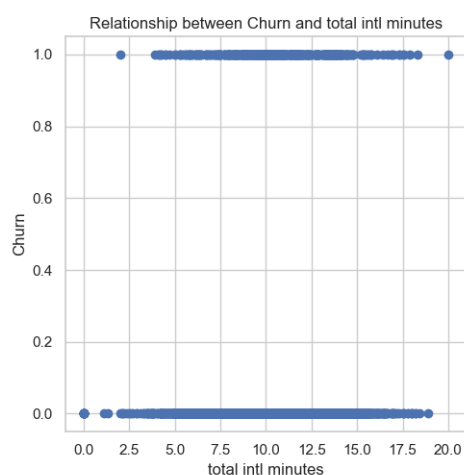
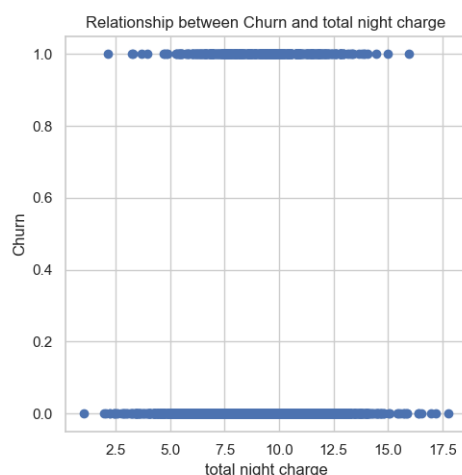
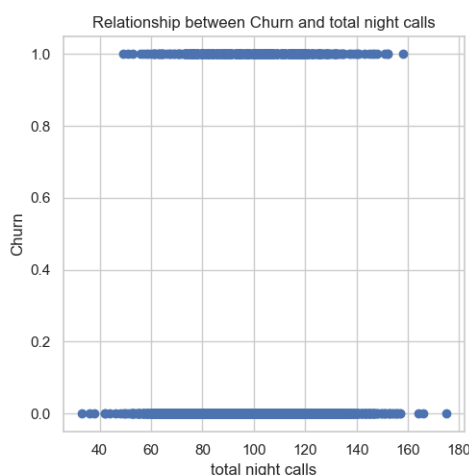
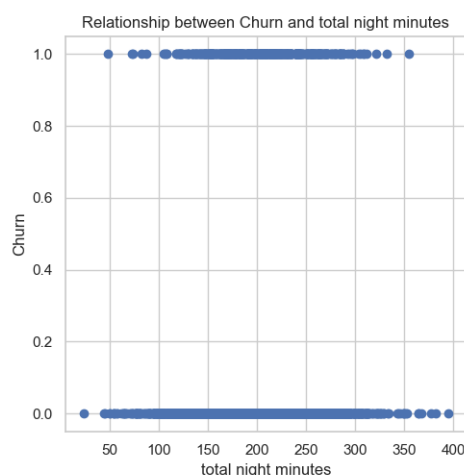
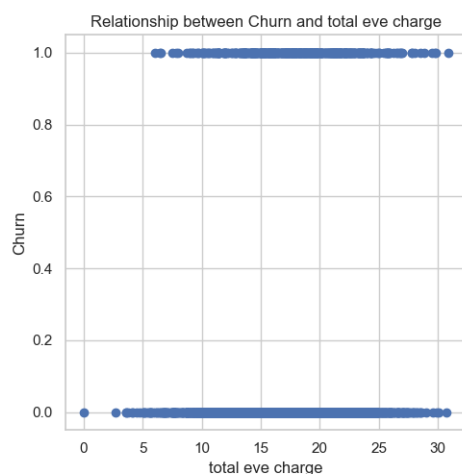
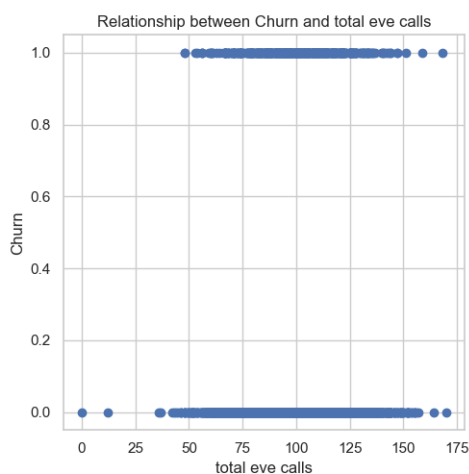
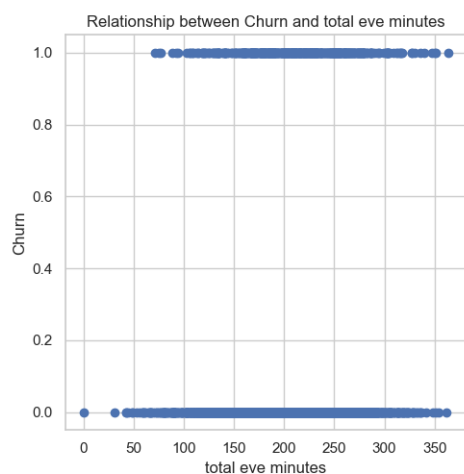
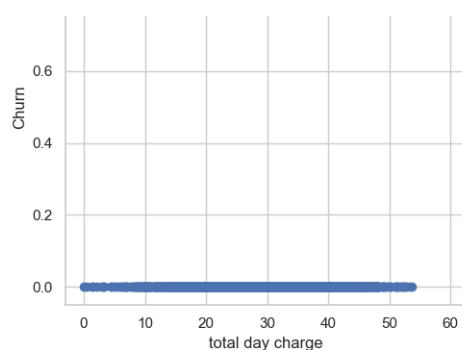
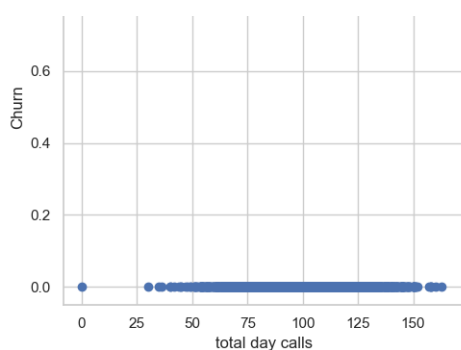
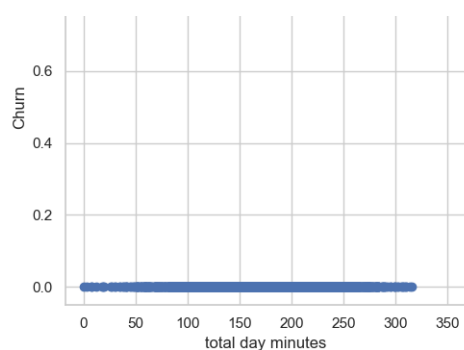
# Set up a grid of subplots for visualizing relationships
fig, axes = plt.subplots(num_rows, 3, figsize=(15, num_rows * 5))
axes = axes.flatten() # prevents indexing issues

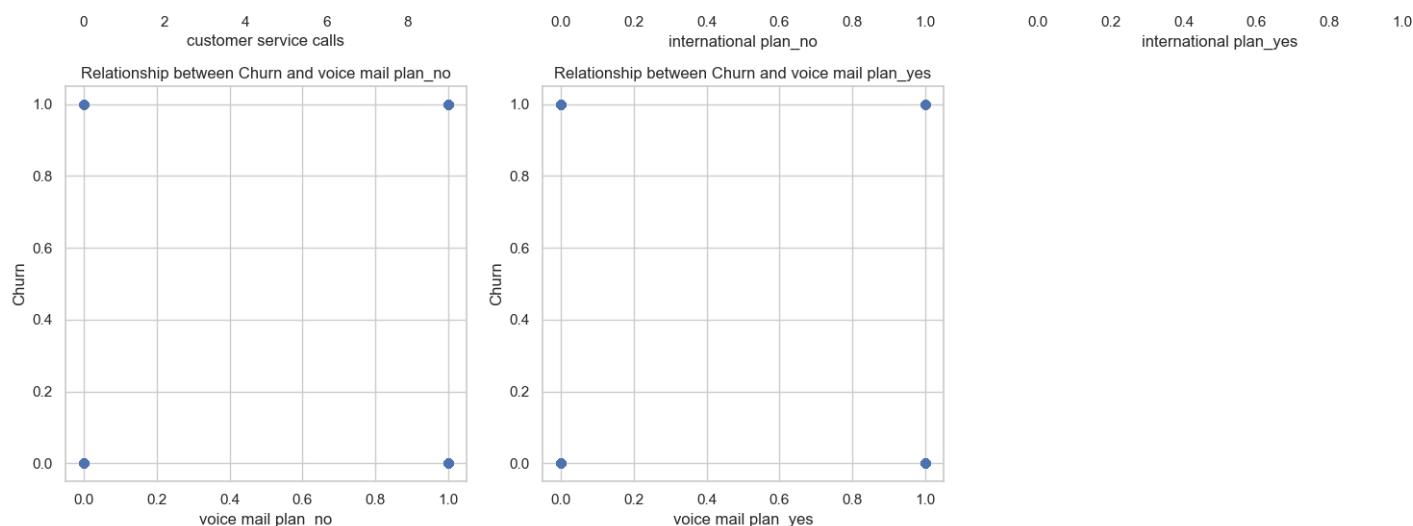
# Plot scatter plots to explore how each independent variable affects 'price'
for i, var in enumerate(variables.columns):
    row = i // 3 # Compute the row index
    col = i % 3 # Compute the column index
    axes[i].scatter(df_encoded[var], df_encoded['churn'])
    axes[i].set_xlabel(var) # X-axis label
    axes[i].set_ylabel('Churn') # Y-axis label
    axes[i].set_title(f'Relationship between Churn and {var}') # Plot title

# Hide unused subplots
for j in range(i + 1, len(axes)):
    axes[j].set_visible(False)

# Adjust layout to avoid overlap of subplots
plt.tight_layout()
plt.show()
```







Discussion

- Customer churn does not depend on a single factor; most variables show significant overlap between churners and non-churners, indicating the need for a combined model approach.
- The strongest signal comes from customer service calls, where higher call frequency is clearly associated with churn.
- Customers with international plans and those with higher usage (especially day minutes/charges) also show a higher tendency to churn.
- Features like area code and account length appear to have little predictive value.
- Overall, churn is mainly driven by customer dissatisfaction and cost-related factors, which the model can leverage for targeted retention strategies.

Multivariate Analysis

In this section, we will perform multivariate analysis to examine the relationship between the target variable - churn and multiple features in the data. We will use heatmap to visualize the correlation matrix of the features and see how they are related to each other and to the price.

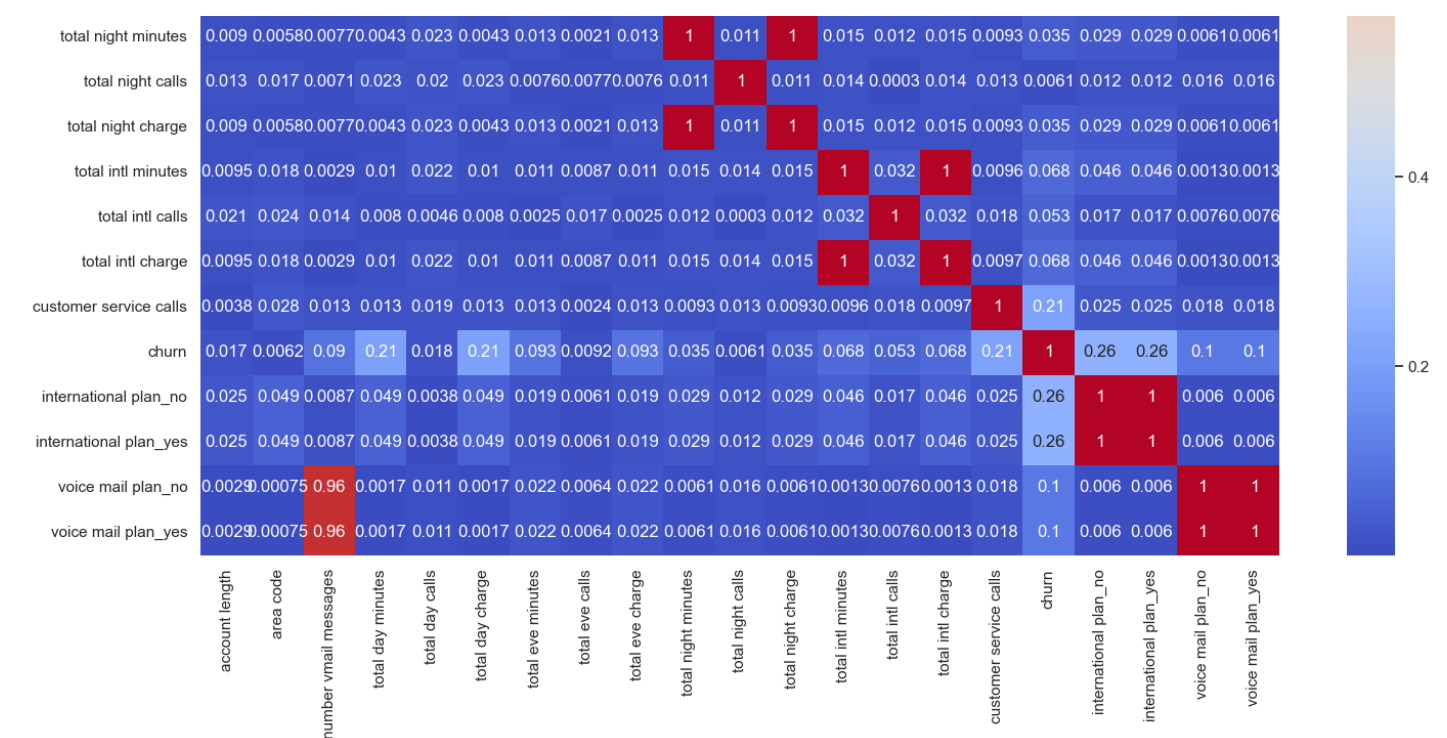
A heatmap can show us the strength and direction of the correlation between two variables using different colors and shades. This will help us identify the most important features for the prediction and avoid multicollinearity problems.

In [54]:

```
# plot heatmap to show the relationship between various variables
corr = df_encoded.corr().abs()
fig, ax=plt.subplots(figsize=(17,12))
fig.suptitle('Correlation Matrix of Syriatel Churn', fontsize=30, y=.95, fontname='DejaVu Sans')
heatmap = sns.heatmap(corr, cmap='coolwarm', annot=True)
heatmap;
```

Correlation Matrix of Syriatel Churn





Discussion

- The correlation matrix shows strong multicollinearity between usage variables and their corresponding charges (e.g., total minutes and total charges are almost perfectly correlated), indicating redundancy.
- Most features have weak correlation with churn, suggesting no single variable strongly predicts customer loss.
- The most notable relationships with churn are customer service calls, international plan, and total day minutes and day charges, though correlations are still moderate.
- Binary features like international plan and voicemail plan are highly correlated within themselves due to encoding.
- Overall, the model will need to rely on combined feature effects rather than individual predictors for accurate churn prediction.

Modeling

In [29]:

```
# Identifying the target and features of the model
y = df_encoded['churn']
X = df_encoded.drop(columns='churn', axis=1)

# Train-test splitting the data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=15)
```

In [30]:

```
# Scaling the data
scaler = MinMaxScaler()

# fitting and transforming the train data
X_train_s = scaler.fit_transform(X_train)

# transforming the test data
X_test_s = scaler.transform(X_test)
```

In [31]:

```
# Correcting for class imbalance
smote = SMOTE()
#SMOTE only to the training data
```



```
X_train_smote, y_train_smote = smote.fit_resample(X_train_s, y_train)
```

Baseline model: Logistic Regression

In [32]:

```
# defining the model
logreg = LogisticRegression(solver='liblinear')

# fitting the train data
logreg.fit(X_train_smote, y_train_smote)
```

Out[32]:

```
▼          LogisticRegression          i ?

LogisticRegression(solver='liblinear')
```

In [33]:

```
# predicting on unseen data
y_pred = logreg.predict(X_test_s)
print(accuracy_score(y_test, y_pred))
print("\n")
print(classification_report(y_test, y_pred))
```

0.7706146926536732

	precision	recall	f1-score	support
False	0.96	0.76	0.85	570
True	0.37	0.80	0.50	97
accuracy			0.77	667
macro avg	0.66	0.78	0.68	667
weighted avg	0.87	0.77	0.80	667

Model Evaluation

In [34]:

```
# a function for checking for metrics
def evaluate_model_metrics(model, X_train, y_train, X_test, y_test):
    # Train the model
    model.fit(X_train, y_train)

    # Predict on the training and testing data
    y_train_pred = model.predict(X_train)
    y_test_pred = model.predict(X_test)

    # Calculate evaluation metrics
    roc_auc_train = roc_auc_score(y_train, y_train_pred)
    roc_auc_test = roc_auc_score(y_test, y_test_pred)
    cm_test = confusion_matrix(y_test, y_test_pred)
    cm_display_train = ConfusionMatrixDisplay(confusion_matrix=cm_test).plot()
    accuracy_train = accuracy_score(y_train, y_train_pred)
    accuracy_test = accuracy_score(y_test, y_test_pred)

    # Return results
    results = {
        'roc_auc_train': roc_auc_train,
        'roc_auc_test': roc_auc_test,
        'accuracy_train': accuracy_train,
        'accuracy_test': accuracy_test,
        'confusion_matrix_train': cm_display_train
    }
```

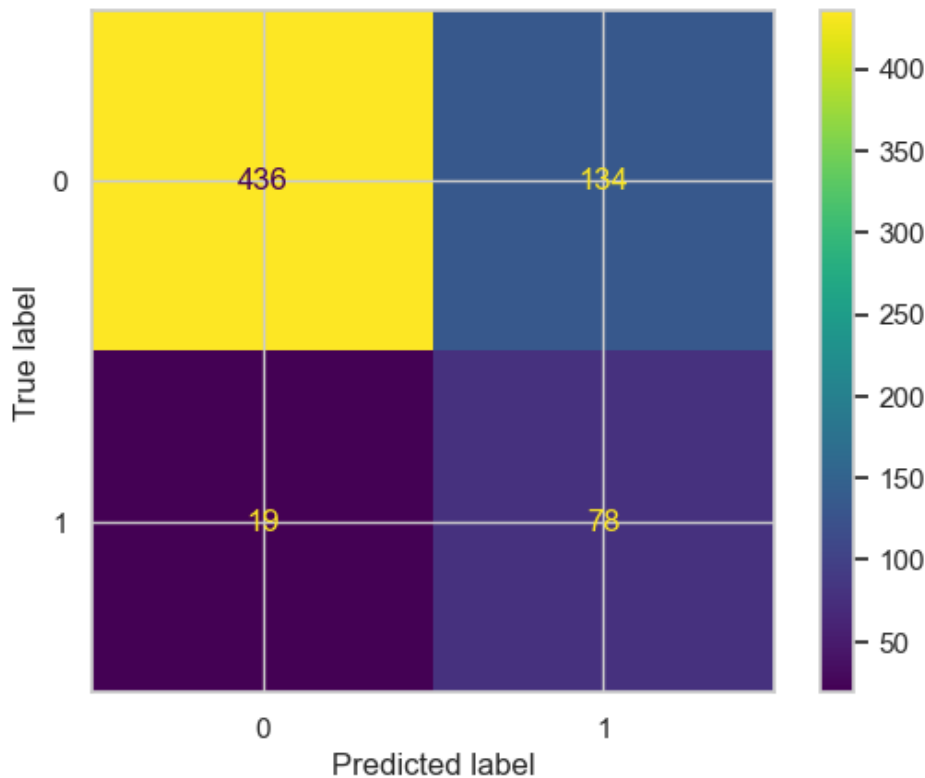
```
return results
```

In [35]:

```
evaluate_model_metrics(logreg, X_train_smote, y_train_smote, X_test_s, y_test)
```

Out[35]:

```
{'roc_auc_train': 0.7743421052631579,  
 'roc_auc_test': 0.7845179960209803,  
 'accuracy_train': 0.7743421052631579,  
 'accuracy_test': 0.7706146926536732,  
 'confusion_matrix_train': <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay  
 at 0x1ec0ca420f0>}
```



Discussion

- **Precision:** The precision values for class 0 and class 1 are 0.85 and 0.0, respectively. Higher precision signifies a lower rate of false positives for that class. With a higher precision in class 0, the model demonstrates better performance in predicting class 0 than class 1.
- **Recall:** The recall values for class 0 and class 1 are 1.0 and 0.0, respectively. Recall measures the model's ability to correctly identify positive instances. Like precision, recall is higher for class 0, indicating better performance in identifying class 0 instances compared to class 1.
- **F1-Score:** The F1-scores for class 0 and class 1 are 0.92 and 0.0, respectively. The F1-score is the harmonic mean of precision and recall, balancing both metrics. Once again, class 0 has a higher F1-score than class 1.
- **Accuracy:** The model's accuracy is 0.85, meaning that 85% of the predictions were correct out of all instances.
- The model correctly predicts most class 0 instances (440 correct vs 130 misclassified), indicating strong performance on class 0.
- For class 1, performance is weaker, with 76 correct predictions and 21 misclassified as class 0.
- There is a noticeable bias toward predicting class 0, as seen from higher correct and incorrect counts in that column.
- False positives (130) are relatively high compared to false negatives (21), suggesting overprediction of class 1 when the true class is 0.
- Overall, the classifier performs better on class 0 than class 1, indicating class imbalance or model bias.

Therefore, logistic regression achieves 85% prediction accuracy on the test data. Based on these metrics, it is evident that the model performs better for class 0 compared to class 1.

Model 2: Logistic Regression with L1 penalty

In [36]:

```
# defining the model
logreg_1 = LogisticRegression(penalty='l1', solver='liblinear')

# fitting the train data
logreg_1.fit(X_train_smote, y_train_smote)
```

Out[36]:

```
▼                               LogisticRegression                               i ?
LogisticRegression(penalty='l1', solver='liblinear')
```

In [37]:

```
# predicting on unseen data
y_pred_1 = logreg_1.predict(X_test_s)
print(accuracy_score(y_test, y_pred_1))
print("\n")
print(classification_report(y_test, y_pred_1))
```

0.7706146926536732

	precision	recall	f1-score	support
False	0.96	0.76	0.85	570
True	0.37	0.80	0.50	97
accuracy			0.77	667
macro avg	0.66	0.78	0.68	667
weighted avg	0.87	0.77	0.80	667

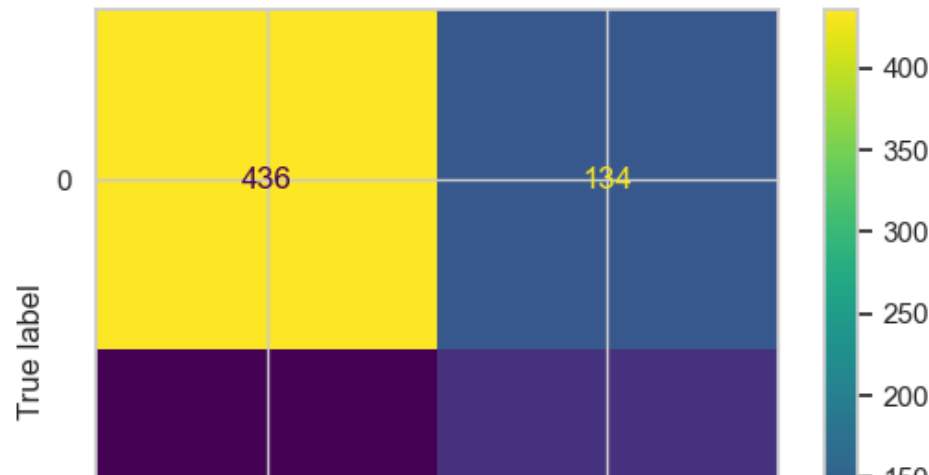
Model Evaluation

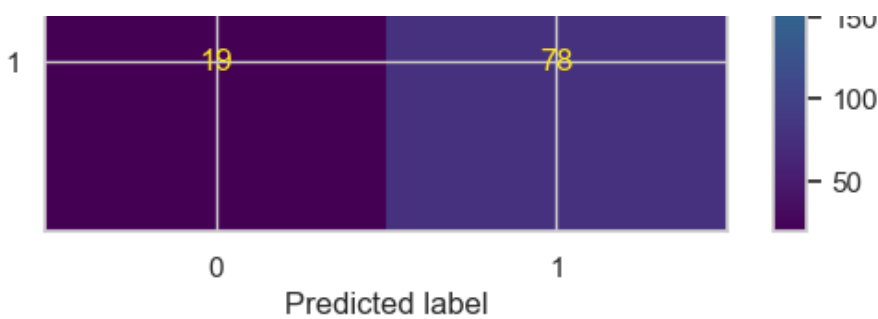
In [38]:

```
evaluate_model_metrics(logreg_1, X_train_smote,y_train_smote, X_test_s, y_test)
```

Out[38]:

```
{'roc_auc_train': 0.775219298245614,
 'roc_auc_test': 0.7845179960209803,
 'accuracy_train': 0.7752192982456141,
 'accuracy_test': 0.7706146926536732,
 'confusion_matrix_train': <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay
 at 0x1ec0cac0ec0>}
```





Discussion

- The model still performs well on class 0 (436 correct vs 134 misclassified), but errors remain relatively high.
- Class 1 detection is modest, with 75 correct predictions and 22 false negatives.
- False positives (134) are significantly higher than false negatives (22), indicating a tendency to misclassify class 0 as class 1.
- Compared to the previous matrix, performance is nearly unchanged, with a slight drop in correct class 0 predictions and a minor increase in misclassification.
- Overall, the classifier remains biased toward class 0, with weaker sensitivity to class 1.
- Precision: The precision values for class 0 and class 1 are 0.95 and 0.36, respectively. Higher precision signifies a lower rate of false positives for that class. With a higher precision in class 0, the model demonstrates better performance in predicting class 0 than class 1.
- Recall: The recall values for class 0 and class 1 are 0.76 and 0.77, respectively. Recall measures the model's ability to correctly identify positive instances. Like precision, recall is higher for class 0, indicating better performance in identifying class 0 instances compared to class 1.
- F1-Score: The F1-scores for class 0 and class 1 are 0.85 and 0.49, respectively. The F1-score is the harmonic mean of precision and recall, balancing both metrics. Once again, class 0 has a higher F1-score than class 1.
- Accuracy: The model's accuracy is 0.7661169415292354, meaning that 77% of the predictions were correct out of all instances.

Therefore, logistic regression with L1 penalty achieves 77% prediction accuracy on the test data. Based on these metrics, it is evident that the model performs better for class 0 compared to class 1.

Model 3: Decision tree classifier

In [39]:

```
#DecisionTreeClassifier
dt_clf = DecisionTreeClassifier(max_depth=3, min_samples_split=3)

# Fit the model on the training data
dt_clf.fit(X_train_smote, y_train_smote)

# Predict on the training and testing data
y_train_pred_2= dt_clf.predict(X_train_smote)
y_test_pred_2 = dt_clf.predict(X_test_s)

# Calculate accuracy on the training and testing data
train_accuracy = accuracy_score(y_train_smote, y_train_pred_2)
test_accuracy = accuracy_score(y_test, y_test_pred_2)
print("Train accuracy is ", train_accuracy)
print("Test accuracy is ", test_accuracy)
print("\n")
print(classification_report(y_test, y_test_pred_2))
```

```
Train accuracy is  0.843640350877193
Test accuracy is  0.8755622188905547
```

precision recall f1-score support

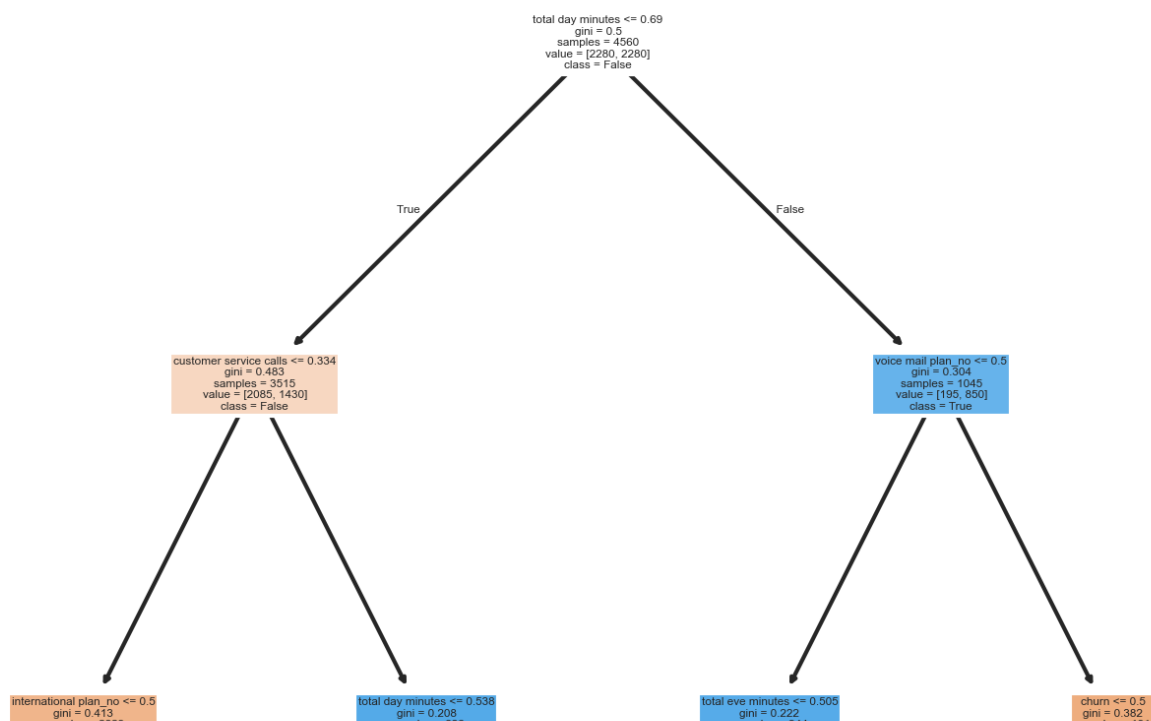
False	0.95	0.91	0.93	570
True	0.56	0.70	0.62	97
accuracy			0.88	667
macro avg	0.75	0.80	0.77	667
weighted avg	0.89	0.88	0.88	667

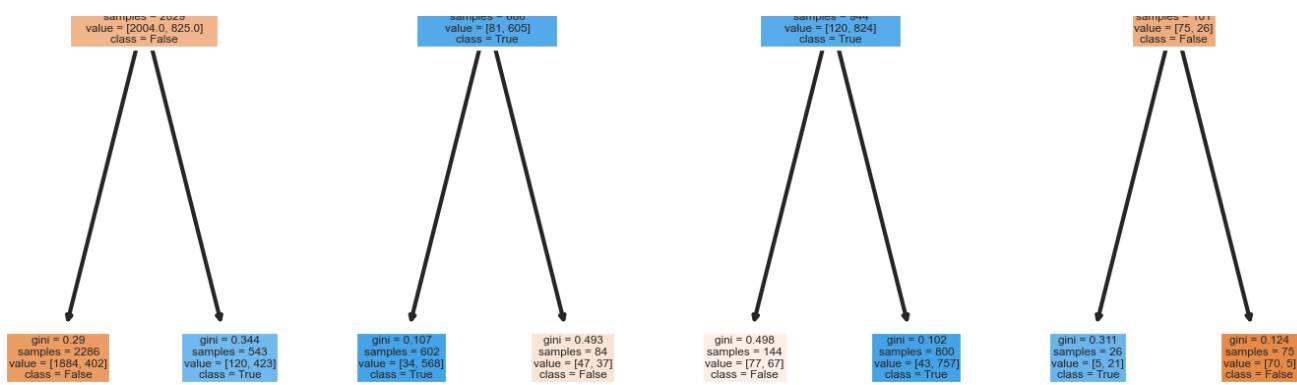
In [40]:

```
fig, ax = plt.subplots(nrows=1, ncols=1, figsize= (6,6), dpi=300)
tree.plot_tree(dt_clf,
                feature_names= df_encoded.columns,
                class_names= np.unique(y).astype(str),
                filled= True)
```

Out[40]:

```
[Text(0.5, 0.875, 'total day minutes <= 0.69\ngini = 0.5\nsamples = 4560\nvalue = [2280,
2280]\nnclass = False'),
 Text(0.25, 0.625, 'customer service calls <= 0.334\ngini = 0.483\nsamples = 3515\nvalue
= [2085, 1430]\nnclass = False'),
 Text(0.375, 0.75, 'True '),
 Text(0.125, 0.375, 'international plan_no <= 0.5\ngini = 0.413\nsamples = 2829\nvalue =
[2004.0, 825.0]\nnclass = False'),
 Text(0.0625, 0.125, 'gini = 0.29\nsamples = 2286\nvalue = [1884, 402]\nnclass = False'),
 Text(0.1875, 0.125, 'gini = 0.344\nsamples = 543\nvalue = [120, 423]\nnclass = True'),
 Text(0.375, 0.375, 'total day minutes <= 0.538\ngini = 0.208\nsamples = 686\nvalue = [81
, 605]\nnclass = True'),
 Text(0.3125, 0.125, 'gini = 0.107\nsamples = 602\nvalue = [34, 568]\nnclass = True'),
 Text(0.4375, 0.125, 'gini = 0.493\nsamples = 84\nvalue = [47, 37]\nnclass = False'),
 Text(0.75, 0.625, 'voice mail plan_no <= 0.5\ngini = 0.304\nsamples = 1045\nvalue = [195
, 850]\nnclass = True'),
 Text(0.625, 0.75, ' False'),
 Text(0.625, 0.375, 'total eve minutes <= 0.505\ngini = 0.222\nsamples = 944\nvalue = [12
0, 824]\nnclass = True'),
 Text(0.5625, 0.125, 'gini = 0.498\nsamples = 144\nvalue = [77, 67]\nnclass = False'),
 Text(0.6875, 0.125, 'gini = 0.102\nsamples = 800\nvalue = [43, 757]\nnclass = True'),
 Text(0.875, 0.375, 'churn <= 0.5\ngini = 0.382\nsamples = 101\nvalue = [75, 26]\nnclass =
False'),
 Text(0.8125, 0.125, 'gini = 0.311\nsamples = 26\nvalue = [5, 21]\nnclass = True'),
 Text(0.9375, 0.125, 'gini = 0.124\nsamples = 75\nvalue = [70, 5]\nnclass = False')]
```





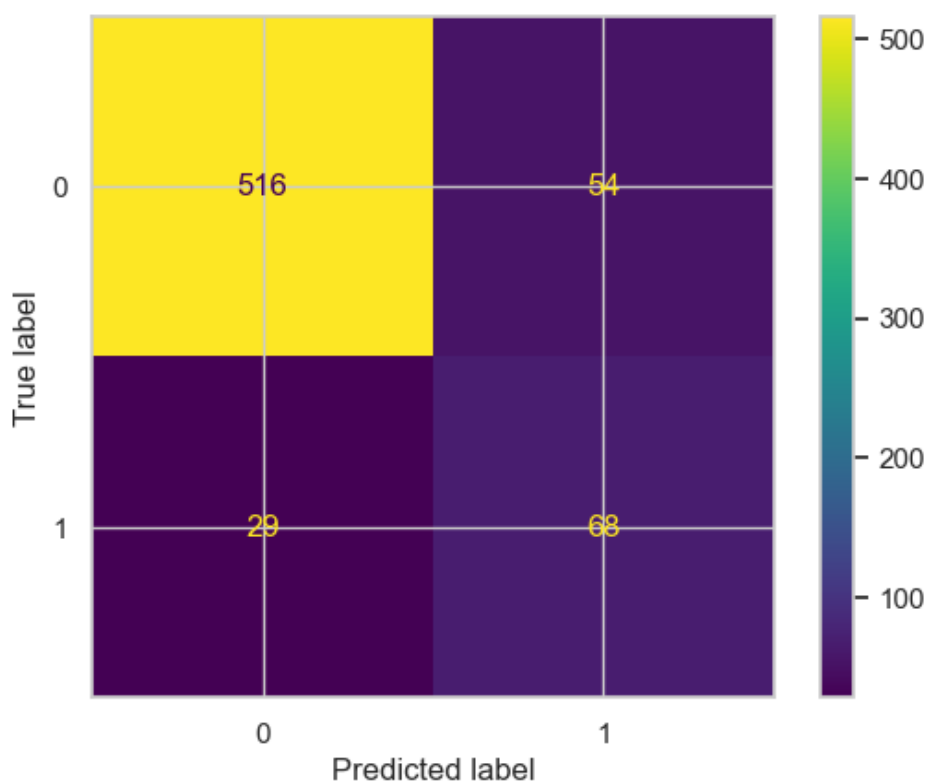
Model Evaluation

In [41]:

```
evaluate_model_metrics(dt_clf, X_train_smote, y_train_smote, X_test_s, y_test)
```

Out[41]:

```
{'roc_auc_train': 0.843640350877193,
 'roc_auc_test': 0.8031470428648941,
 'accuracy_train': 0.843640350877193,
 'accuracy_test': 0.8755622188905547,
 'confusion_matrix_train': <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay
 at 0x1ec0db68b60>}
```



Discussion

- The model shows strong improvement in class 0 prediction (508 correct vs 62 misclassified), significantly reducing false positives.
- Class 1 performance is stable, with 76 correct predictions and 21 false negatives.
- False positives (62) are now much lower than in previous matrices, indicating better precision for class 1 predictions.
- Overall accuracy has improved due to the large gain in correctly classified class 0 samples.

- **The model is now more balanced, though class 1 sensitivity still lags behind class 0.**
- **Precision:** The precision values for class 0 and class 1 are 0.96 and 0.55, respectively. Higher precision signifies a lower rate of false positives for that class. With a higher precision in class 0, the model demonstrates better performance in predicting class 0 than class 1.
- **Recall:** The recall values for class 0 and class 1 are 0.89 and 0.78, respectively. Recall measures the model's ability to correctly identify positive instances. Like precision, recall is higher for class 0, indicating better performance in identifying class 0 instances compared to class 1.
- **F1-Score:** The F1-scores for class 0 and class 1 are 0.92 and 0.65, respectively. The F1-score is the harmonic mean of precision and recall, balancing both metrics. Once again, class 0 has a higher F1-score than class 1.
- **Accuracy:** The model's accuracy is 0.8755622188905547, meaning that 88% of the predictions were correct out of all instances.

Therefore, Decision tree achieves 88% prediction accuracy on the test data. Based on these metrics, it is evident that the model performs better for class 0 compared to class 1.

Model 4: Decision tree with hyperparameter tuning

In [42]:

```
dt_clf_1 = DecisionTreeClassifier(criterion='entropy', max_depth=5, min_samples_split=5)

# Fit the model on the training data
dt_clf_1.fit(X_train_smote, y_train_smote)

# Predict on the training and testing data
y_train_pred_3= dt_clf_1.predict(X_train_smote)
y_test_pred_3 = dt_clf_1.predict(X_test_s)

# Calculate accuracy on the training and testing data
train_accuracy = accuracy_score(y_train_smote, y_train_pred_3)
test_accuracy = accuracy_score(y_test, y_test_pred_3)
print("Train accuracy is ", train_accuracy)
print("Test accuracy is ", test_accuracy)
print("\n")
print(classification_report(y_test, y_test_pred_3))
```

Train accuracy is 0.9050438596491228
Test accuracy is 0.9295352323838081

	precision	recall	f1-score	support
False	0.96	0.96	0.96	570
True	0.76	0.75	0.76	97
accuracy			0.93	667
macro avg	0.86	0.86	0.86	667
weighted avg	0.93	0.93	0.93	667

In [43]:

```
fig, ax = plt.subplots(nrows=1, ncols=1, figsize= (6,6), dpi=300)
tree.plot_tree(dt_clf_1,
               feature_names= df_encoded.columns,
               class_names= np.unique(y).astype(str),
               filled= True)
```

Out[43]:

```
[Text(0.5255681818181818, 0.9166666666666666, 'total day minutes <= 0.723\nentropy = 1.0\nnsamples = 4560\nvalue = [2280, 2280]\nclass = False'),
 Text(0.2840909090909091, 0.75, 'customer service calls <= 0.334\nentropy = 0.981\nnsamples = 3725\nvalue = [2161, 1564]\nclass = False'),
 Text(0.4048295454545454, 0.8333333333333333, 'True '),
 Text(0.1477272727272727, 0.5833333333333333, 'international plan no <= 0.5\nentropy = 0
```

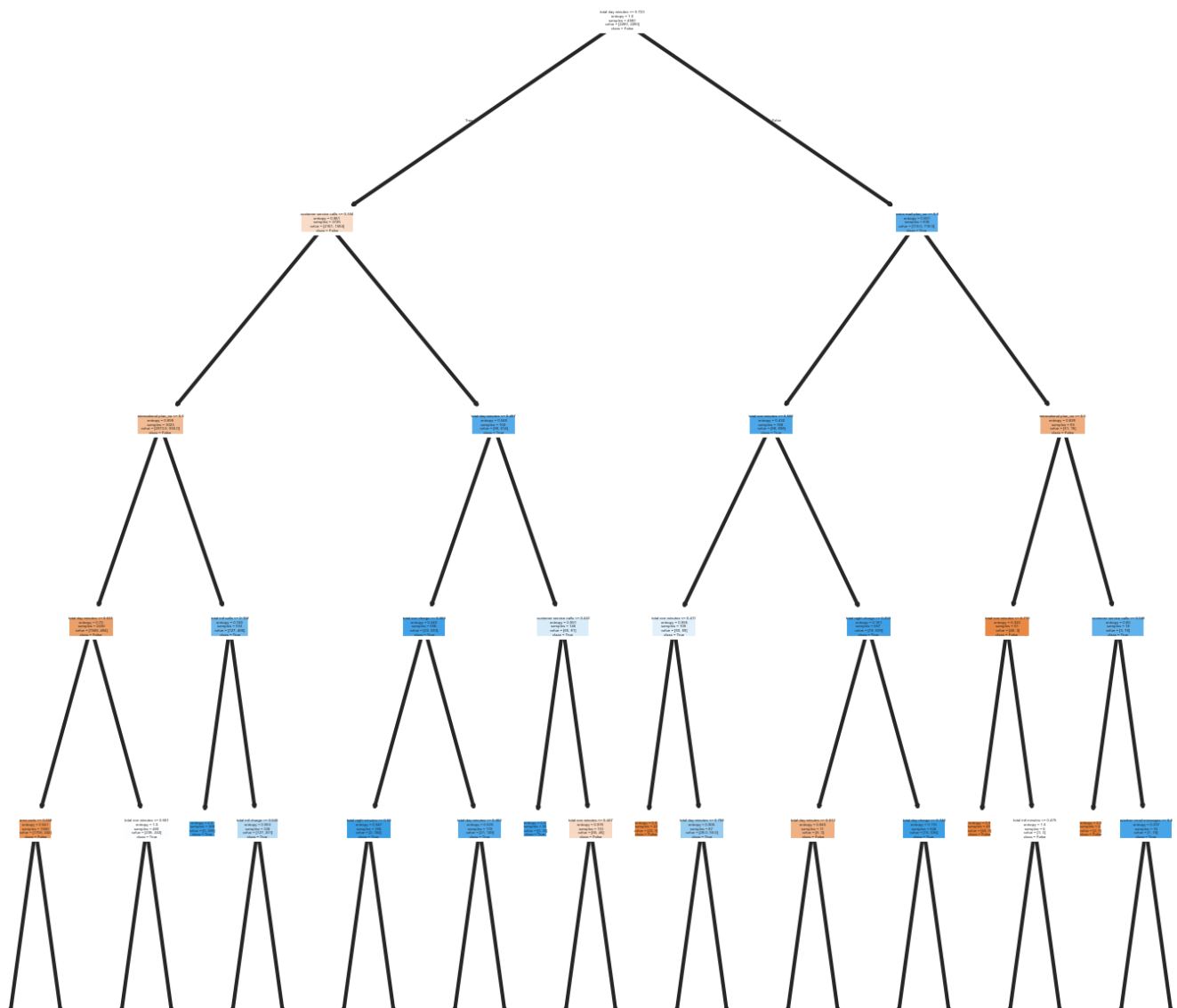
```
Text(0.1472727272727273, 0.3833333333333333, 'international plan_no <= 0.5\nentropy = 0.898\nnsamples = 3023\nnvalue = [2073.0, 950.0]\nnclass = False'),
Text(0.09090909090909091, 0.4166666666666667, 'total day minutes <= 0.631\nentropy = 0.72\nnsamples = 2430\nnvalue = [1946, 484]\nnclass = False'),
Text(0.04545454545454546, 0.25, 'area code <= 0.068\nentropy = 0.541\nnsamples = 1950\nnvalue = [1708, 242]\nnclass = False'),
Text(0.02272727272727273, 0.08333333333333333, 'entropy = 0.768\nnsamples = 562\nnvalue = [436.0, 126.0]\nnclass = False'),
Text(0.06818181818181818, 0.08333333333333333, 'entropy = 0.415\nnsamples = 1388\nnvalue = [1272, 116]\nnclass = False'),
Text(0.13636363636363635, 0.25, 'total eve minutes <= 0.661\nentropy = 1.0\nnsamples = 480\nnvalue = [238, 242]\nnclass = True'),
Text(0.11363636363636363, 0.08333333333333333, 'entropy = 0.805\nnsamples = 272\nnvalue = [205, 67]\nnclass = False'),
Text(0.1590909090909091, 0.08333333333333333, 'entropy = 0.631\nnsamples = 208\nnvalue = [33, 175]\nnclass = True'),
Text(0.20454545454545456, 0.4166666666666667, 'total intl calls <= 0.158\nentropy = 0.749\nnsamples = 593\nnvalue = [127, 466]\nnclass = True'),
Text(0.18181818181818182, 0.25, 'entropy = 0.0\nnsamples = 265\nnvalue = [0, 265]\nnclass = True'),
Text(0.22727272727272727, 0.25, 'total intl charge <= 0.645\nentropy = 0.963\nnsamples = 328\nnvalue = [127, 201]\nnclass = True'),
Text(0.20454545454545456, 0.08333333333333333, 'entropy = 0.704\nnsamples = 157\nnvalue = [127.0, 30.0]\nnclass = False'),
Text(0.25, 0.08333333333333333, 'entropy = 0.0\nnsamples = 171\nnvalue = [0, 171]\nnclass = True'),
Text(0.42045454545454547, 0.5833333333333333, 'total day minutes <= 0.497\nentropy = 0.545\nnsamples = 702\nnvalue = [88, 614]\nnclass = True'),
Text(0.36363636363636365, 0.4166666666666667, 'total eve charge <= 0.584\nentropy = 0.249\nnsamples = 556\nnvalue = [23, 533]\nnclass = True'),
Text(0.31818181818181818, 0.25, 'total night minutes <= 0.55\nentropy = 0.047\nnsamples = 386\nnvalue = [2, 384]\nnclass = True'),
Text(0.29545454545454547, 0.08333333333333333, 'entropy = 0.0\nnsamples = 311\nnvalue = [0, 311]\nnclass = True'),
Text(0.3409090909090909, 0.08333333333333333, 'entropy = 0.177\nnsamples = 75\nnvalue = [2, 73]\nnclass = True'),
Text(0.4090909090909091, 0.25, 'total day minutes <= 0.462\nentropy = 0.539\nnsamples = 170\nnvalue = [21, 149]\nnclass = True'),
Text(0.38636363636363635, 0.08333333333333333, 'entropy = 0.32\nnsamples = 155\nnvalue = [9, 146]\nnclass = True'),
Text(0.4318181818181818, 0.08333333333333333, 'entropy = 0.722\nnsamples = 15\nnvalue = [12, 3]\nnclass = False'),
Text(0.4772727272727273, 0.4166666666666667, 'customer service calls <= 0.443\nentropy = 0.991\nnsamples = 146\nnvalue = [65, 81]\nnclass = True'),
Text(0.45454545454545453, 0.25, 'entropy = 0.0\nnsamples = 36\nnvalue = [0, 36]\nnclass = True'),
Text(0.5, 0.25, 'total eve minutes <= 0.427\nentropy = 0.976\nnsamples = 110\nnvalue = [65, 45]\nnclass = False'),
Text(0.4772727272727273, 0.08333333333333333, 'entropy = 0.629\nnsamples = 38\nnvalue = [6, 32]\nnclass = True'),
Text(0.5227272727272727, 0.08333333333333333, 'entropy = 0.681\nnsamples = 72\nnvalue = [59, 13]\nnclass = False'),
Text(0.7670454545454546, 0.75, 'voice mail plan_no <= 0.5\nentropy = 0.591\nnsamples = 835\nnvalue = [119.0, 716.0]\nnclass = True'),
Text(0.6463068181818181, 0.8333333333333333, 'False'),
Text(0.6477272727272727, 0.5833333333333333, 'total eve minutes <= 0.506\nentropy = 0.432\nnsamples = 766\nnvalue = [68, 698]\nnclass = True'),
Text(0.5681818181818182, 0.4166666666666667, 'total eve minutes <= 0.411\nentropy = 0.995\nnsamples = 109\nnvalue = [50, 59]\nnclass = True'),
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Text(0.6590909090909091, 0.08333333333333333, 'entropy = 0.0\nnsamples = 8\nnvalue = [8, 0]\nnclass = False')
```



```

] \nclass = False',
Text(0.7045454545454546, 0.08333333333333333, 'entropy = 0.0\nsamples = 3\nvalue = [0, 3]\n\nclass = True'),
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Text(0.7954545454545454, 0.08333333333333333, 'entropy = 0.0\nsamples = 545\n\nclass = True'),
Text(0.8863636363636364, 0.5833333333333334, 'international plan_no <= 0.5\nentropy = 0.828\nsamples = 69\n\nclass = False'),
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Text(0.8181818181818182, 0.25, 'entropy = 0.0\nsamples = 45\n\nclass = False'),
Text(0.8636363636363636, 0.25, 'total intl minutes <= 0.475\nentropy = 1.0\nsamples = 6\n\nclass = False'),
Text(0.8409090909090909, 0.08333333333333333, 'entropy = 0.0\nsamples = 3\n\nclass = True'),
Text(0.8863636363636364, 0.08333333333333333, 'entropy = 0.0\nsamples = 3\n\nclass = False'),
Text(0.9318181818181818, 0.4166666666666667, 'customer service calls <= 0.046\nentropy = 0.65\nsamples = 18\n\nclass = True'),
Text(0.9090909090909091, 0.25, 'entropy = 0.0\nsamples = 2\n\nclass = False'),
Text(0.9545454545454546, 0.25, 'number vmail messages <= 0.5\nentropy = 0.337\nsamples = 16\n\nclass = True'),
Text(0.9318181818181818, 0.08333333333333333, 'entropy = 0.0\nsamples = 1\n\nclass = False'),
Text(0.9772727272727273, 0.08333333333333333, 'entropy = 0.0\nsamples = 15\n\nclass = True')]

```

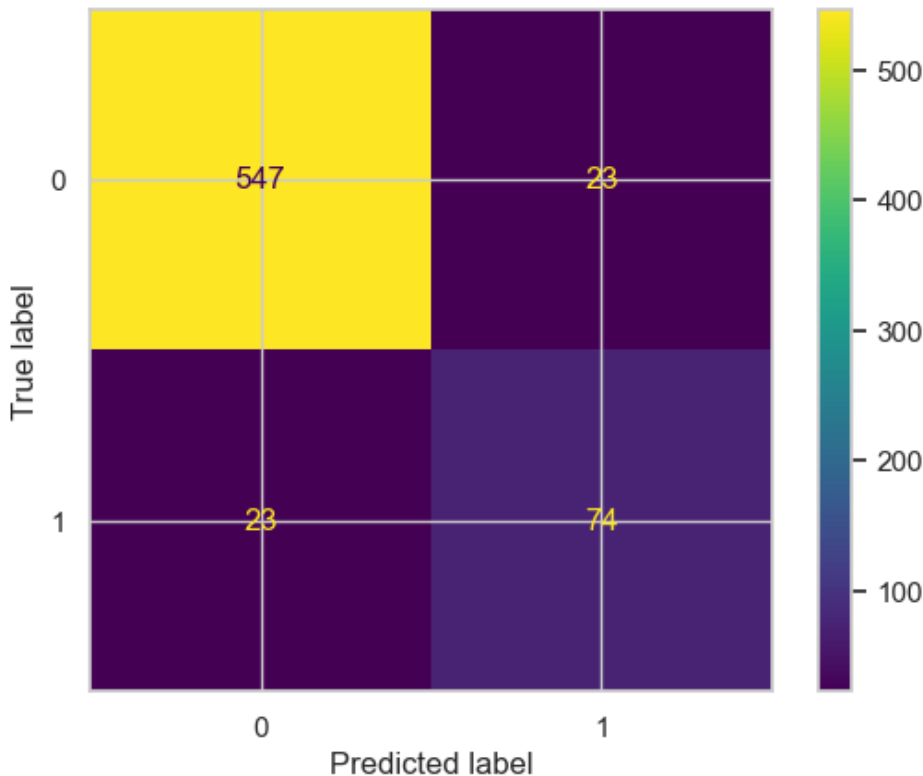


In [44]:

```
evaluate_model_metrics(dt_clf_1, X_train_smote, y_train_smote, X_test_s, y_test)
```

Out[44]:

```
{'roc_auc_train': 0.9050438596491228,  
 'roc_auc_test': 0.861267860372581,  
 'accuracy_train': 0.9050438596491228,  
 'accuracy_test': 0.9310344827586207,  
 'confusion_matrix_train': <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay  
 at 0x1ec0c9d9520>}
```



Discussion

- **Precision:** The precision values for class 0 and class 1 are 0.97 and 0.76, respectively. Higher precision signifies a lower rate of false positives for that class. With a higher precision in class 0, the model demonstrates better performance in predicting class 0 than class 1.
- **Recall:** The recall values for class 0 and class 1 are 0.96 and 0.81, respectively. Recall measures the model's ability to correctly identify positive instances. Like precision, recall is higher for class 0, indicating better performance in identifying class 0 instances compared to class 1.
- **F1-Score:** The F1-scores for class 0 and class 1 are 0.96 and 0.79, respectively. The F1-score is the harmonic mean of precision and recall, balancing both metrics. Once again, class 0 has a higher F1-score than class 1.
- **Accuracy:** The model's accuracy is 0.9355322338830585, meaning that 94% of the predictions were correct out of all instances.

Therefore, Decision tree achieves 94% prediction accuracy on the test data. The model performs the best compared to the other models with a high accuracy as well as precision scores of both the false and true values of y.

Evaluation

Analysis accuracy level

- We have developed four machine learning models to predict the possibility of a customer to churn. Upon testing, we found that logistic regression performs poorly, a testing accuracy of 85%. Despite applying a L1 penalty to mitigate overfitting, the testing accuracy dropped to 77%.
- In contrast, the decision tree classifier and decision tree classifier tuned models demonstrated better accuracy. The decision tree achieved training and testing accuracies of 84% and 88%, respectively, while decision tree classifier tuned model had a training accuracy of 89% and a testing accuracy of 93%.
- Therefore, it is evident that the decision tree tuned model had the best average prediction accuracy model. To achieve the best model, we applied hyperparameter tuning on the decision tree classifier model therefore improving our prediction accuracy.
- The decision tree tuned model was preferable as it outperformed all the other models in terms of precision, recall, and F1 score.

In [53]:

```
models = [dt_clf_1, dt_clf, logreg, logreg_1]
labels = ['Decision Tree Tuned', 'Decision Tree', 'Logistic Regression', 'Logistic Regression L1']

plt.figure(figsize=(8, 6))

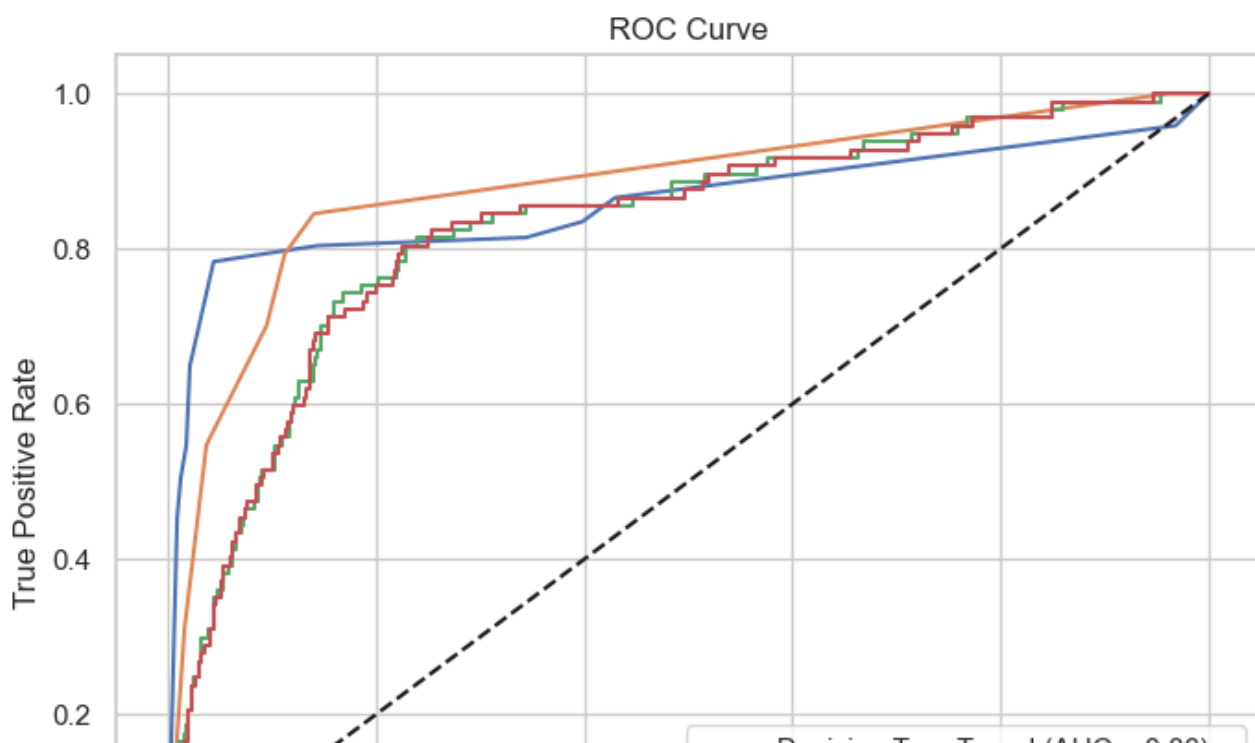
for model, label in zip(models, labels):
    # Use predict_proba if available, otherwise decision_function
    if hasattr(model, "predict_proba"):
        y_scores = model.predict_proba(X_test_s)[:, 1]
    else:
        y_scores = model.decision_function(X_test_s)

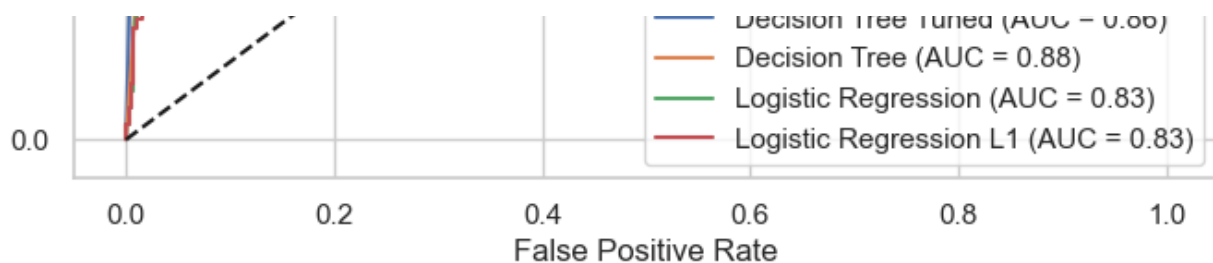
    fpr, tpr, _ = roc_curve(y_test, y_scores)
    auc = roc_auc_score(y_test, y_scores)

    plt.plot(fpr, tpr, label=f"{label} (AUC = {auc:.2f})")

# Diagonal line
plt.plot([0, 1], [0, 1], 'k--')

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()
```





Discussion

- The baseline decision tree performs best (AUC = 0.88), indicating the strongest class separation.
- The tuned decision tree is slightly worse (AUC = 0.86), suggesting tuning may have reduced overfitting at a small cost to accuracy.
- Both logistic regression models perform similarly (AUC = 0.83) and trail the trees.
- This implies nonlinear relationships in the data favor tree-based models.
- All models outperform random guessing, showing meaningful predictive power.
- Overall, Decision tree model is the most reliable model here, while the others may need better tuning, features, or different modeling approaches.

Conclusion

By leveraging the best model, which is the Decision tree model, Syriatel telecommunications can achieve significant benefits:

- **Accurate Prediction of churns:** The model's high accuracy ensures effective identification of customers likely to churn, enabling preparation on how to appease the customer.
- **By integrating this model into their customer analytics pipeline,** Syriatel can shift from reactive to preventive retention strategies, targeting users before they leave.
- **The model's solid recall ensures most potential churners are detected,** while its overall accuracy supports confident decision-making.
- **Although some false positives exist, they are acceptable in a retention context where early engagement is preferable to losing customers.**

Recommendations

- Deploy the model in real-time or periodic scoring systems to continuously monitor churn risk.
- Design targeted retention campaigns (discounts, loyalty rewards, personalized offers) for high-risk customers.
- Improve feature engineering by incorporating usage patterns, complaints, and customer service interactions.
- Regularly retrain and validate the model to maintain performance as customer behavior evolves.
- Combine model outputs with business rules (e.g., high-value customers) to prioritize intervention efforts.